Product, Quotient, and Trigonometric Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Product Rule**

A differentiable function $f(t)$ is represented by the graph below. What is the total change in $f(t)$ from $t = 2$ to $t = 5$?

The total change in a function $f(t)$ from $t = a$ to $t = b$ is given by the definite integral $\int_a^b f'(t)\,dt$. Alternatively, if the graph represents the function $f(t)$ itself, the total change is simply $f(b) - f(a)$. From the graph, we can observe that $f(2) = 1$ and $f(5) = 2$. Therefore, the total change is $f(5) - f(2) = 2 - 1 = 1$.

2 **[Foundational]** • **Target: The Quotient Rule**

Given $\int_0^1 1x^2\,dx = \frac{1}{3}$ and $\int_1^3 1x^2\,dx = \frac{26}{3}$, find $\int_0^3 1x^2\,dx$.

By additivity of the definite integral: $\int_0^3 = \int_0^1 + \int_1^3 = \frac{1}{3} + \frac{26}{3} = 9$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

3 **[Foundational]** • **Target: Derivatives of Basic Trigonometric Functions**

Consider the function $f(x) = 3x^2 - 6x + 5$. What is the instantaneous rate of change of $f(x)$ at $x = 2$?

The instantaneous rate of change of a function at a point is given by its derivative evaluated at that point. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(3x^2 - 6x + 5) = 6x - 6$. Next, evaluate $f'(x)$ at $x = 2$: $f'(2) = 6(2) - 6 = 12 - 6 = 6$. Therefore, the instantaneous rate of change of $f(x)$ at $x = 2$ is 6.

4 **[Foundational]** • **Target: Higher-Order Derivatives & Acceleration**

A conical tank has height 15 ft and radius 7 ft at the top (radius:height ratio constant as it drains). Water drains at 6 ft^3/min. Find $\frac{dh}{dt}$ (in ft/min) when the water depth is 13 ft.

Since $r = \frac{7}{15}h$, $V = \frac{1}{3}\pi\left(\frac{7}{15}\right)^2 h^3$. Differentiating and solving for $\frac{dh}{dt}$ at $h = 13$ gives $-\frac{1350}{8281}/\pi$ ft/min.

5 **[Foundational]** • **Target: The Product Rule**

Product rule for $\frac{d}{dx}[f(x)g(x)]$:.

Final Answer: $f'g + fg'$. Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, Product rule: $f'g + fg'$ Therefore, the verified correct option is $f'g + fg'$.

6 **[Foundational]** • **Target: The Quotient Rule**

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{4x+2}$.

Final Answer: $\frac{1}{2(2x+1)^2}$. Quotient rule: $f'(x) = \frac{1 \cdot (4x+2) - x \cdot 4}{(4x+2)^2} = \frac{1}{2(2x+1)^2}$.

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7 [Foundational] • Target: Derivatives of Basic Trigonometric Functions

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{5x+3}$.

Final Answer: $\frac{3}{(5x+3)^2}$. Quotient rule: $f'(x) = \frac{1 \cdot (5x+3) - x \cdot 5}{(5x+3)^2} = \frac{3}{(5x+3)^2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{2x+2}$.

Final Answer: $\frac{1}{2(x+1)^2}$. Quotient rule: $f'(x) = \frac{1 \cdot (2x+2) - x \cdot 2}{(2x+2)^2} = \frac{1}{2(x+1)^2}$.

9 [Intermediate] • Target: The Product Rule

Use the product rule to find $f'(x)$ if $f(x) = x^4(3x-4)$.

Final Answer: $x^3(15x-16)$. Product rule: $f'(x) = 4x^3(3x-4) + x^4(3) = x^3(15x-16)$.

10 [Intermediate] • Target: The Quotient Rule

Use the product rule to find $f'(x)$ if $f(x) = x^4(2x+3)$.

Final Answer: $x^3(10x+12)$. Product rule: $f'(x) = 4x^3(2x+3) + x^4(2) = x^3(10x+12)$.

11 [Intermediate] • Target: Derivatives of Basic Trigonometric Functions

Use the product rule to find $f'(x)$ if $f(x) = x^4(4x-4)$.

Final Answer: $x^3(20x-16)$. Product rule: $f'(x) = 4x^3(4x-4) + x^4(4) = x^3(20x-16)$.

12 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Use the product rule to find $f'(x)$ if $f(x) = x^3(5x-1)$.

Final Answer: $x^2(20x-3)$. Product rule: $f'(x) = 3x^2(5x-1) + x^3(5) = x^2(20x-3)$.

13 [Intermediate] • Target: The Product Rule

Use the product rule to find $f'(x)$ if $f(x) = x^3(5x+3)$.

Final Answer: $x^2(20x+9)$. Product rule: $f'(x) = 3x^2(5x+3) + x^3(5) = x^2(20x+9)$.

14 [Intermediate] • Target: The Quotient Rule

Use the product rule to find $f'(x)$ if $f(x) = x^3(4x-1)$.

Final Answer: $x^2(16x-3)$. Product rule: $f'(x) = 3x^2(4x-1) + x^3(4) = x^2(16x-3)$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 3.01 The Chain Rule

Use the product rule to find $f'(x)$ if $f(x) = x^4(4x + 1)$.

Final Answer: $x^3(20x + 4)$. Product rule: $f'(x) = 4x^3(4x + 1) + x^4(4) = x^3(20x + 4)$.

16 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Use the product rule to find $f'(x)$ if $f(x) = x^4(5x - 1)$.

Final Answer: $x^3(25x - 4)$. Product rule: $f'(x) = 4x^3(5x - 1) + x^4(5) = x^3(25x - 4)$.

17 [Challenging] • Target: The Product Rule • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Use the product rule to find $f'(x)$ if $f(x) = x^3(2x + 1)$.

Final Answer: $x^2(8x + 3)$. Product rule: $f'(x) = 3x^2(2x + 1) + x^3(2) = x^2(8x + 3)$.

18 [Challenging] • Target: The Quotient Rule • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Use the product rule to find $f'(x)$ if $f(x) = x^3(1x + 2)$.

Final Answer: $x^2(4x + 6)$. Product rule: $f'(x) = 3x^2(1x + 2) + x^3(1) = x^2(4x + 6)$.

19 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use the product rule to find $f'(x)$ if $f(x) = x^3(2x + 3)$.

Final Answer: $x^2(8x + 9)$. Product rule: $f'(x) = 3x^2(2x + 3) + x^3(2) = x^2(8x + 9)$.

20 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use the product rule to find $f'(x)$ if $f(x) = x^2(4x + 2)$.

Final Answer: $4x(3x + 1)$. Product rule: $f'(x) = 2x^1(4x + 2) + x^2(4) = 4x(3x + 1)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Product Rule

Find the derivative of $f(x) = x^3 \sin x$.

a. $3x^2 \sin x + x^3 \cos x$ ✓

b. $3x^2 \cos x$

c. $3x^2 \sin x - x^3 \cos x$

d. $x^2(3\sin x + \cos x)$

Product rule: $\frac{d}{dx}[uv] = u'v + uv'$. Here $3x^2 \sin x + x^3 \cos x$.

22 [Foundational] • Target: The Product Rule

If $h(x) = f(x)g(x)$ with $f(2) = 3, f'(2) = -1, g(2) = 4,$ and $g'(2) = 5$, what is $h'(2)$?

a. 11 ✓

b. 19

c. -14

d. 15

$h'(2) = f'(2)g(2) + f(2)g'(2) = (-1)(4) + (3)(5) = -4 + 15 = 11$.

23 [Foundational] • Target: The Product Rule

Find $\frac{d}{dx}[e^x \cos x]$.

a. $e^x(\cos x - \sin x)$ ✓

b. $e^x(\cos x + \sin x)$

c. $-e^x \sin x$

d. $e^x \cos x$

Product rule: $e^x \cos x + e^x(-\sin x) = e^x(\cos x - \sin x)$.

24 [Foundational] • Target: The Quotient Rule

Differentiate $y = \frac{2x-3}{x+4}$.

a. $\frac{11}{(x+4)^2}$ ✓

b. $\frac{5}{(x+4)^2}$

c. $\frac{2x+5}{(x+4)^2}$

d. $\frac{2}{(x+4)^2}$

Quotient rule: $\frac{2(x+4)-(2x-3)(1)}{(x+4)^2} = \frac{2x+8-2x+3}{(x+4)^2} = \frac{11}{(x+4)^2}$.

25 [Foundational] • Target: The Quotient Rule

Let $k(x) = \frac{f(x)}{g(x)}$. If $f(1) = 6, f'(1) = 2, g(1) = 3,$ and $g'(1) = -1$, find $k'(1)$.

a. $\frac{4}{3}$ ✓

b. 0

c. 2

d. $-\frac{4}{3}$

$k'(1) = \frac{f'(1)g(1) - f(1)g'(1)}{[g(1)]^2} = \frac{2(3) - 6(-1)}{3^2} = \frac{6+6}{9} = \frac{12}{9} = \frac{4}{3}$.

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26 [Foundational] • Target: The Quotient Rule

Find the derivative of $f(x) = \frac{e^x}{x^2}$.

a. $\frac{e^x(x-2)}{x^3}$ ✓

b. $\frac{e^x(x+2)}{x^3}$

c. $\frac{e^x}{2x}$

d. $\frac{e^x(x-1)}{x^3}$

Quotient rule: $\frac{e^x(x^2)-e^x(2x)}{x^4} = \frac{xe^x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$.

27 [Foundational] • Target: Derivatives of Basic Trigonometric Functions

What is $\frac{d}{dx}[\tan x]$?

a. $\sec^2 x$ ✓

b. $\sec x \tan x$

c. $-\csc^2 x$

d. $\cot x$

By definition, $\frac{d}{dx}[\frac{\sin x}{\cos x}] = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$.

28 [Foundational] • Target: Derivatives of Basic Trigonometric Functions

What is $\frac{d}{dx}[\sec x]$?

a. $\sec x \tan x$ ✓

b. $\sec^2 x$

c. $-\csc x \cot x$

d. $\tan^2 x$

By the quotient rule on $\frac{1}{\cos x}$: $\frac{0-(-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \tan x$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Derivatives of Basic Trigonometric Functions

What is $\frac{d}{dx}[\cot x]$?

a. $-\csc^2 x$ ✓

b. $\csc^2 x$

c. $-\sec^2 x$

d. $-\csc x \cot x$

$\frac{d}{dx}[\cot x] = -\csc^2 x$.

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30 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Find the second derivative $f''(x)$ of $f(x) = \sin x$.

a. $-\sin x$ ✓

b. $\sin x$

c. $-\cos x$

d. $\cos x$

$f'(x) = \cos x$. Then $f''(x) = \frac{d}{dx}[\cos x] = -\sin x$.

31 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

If position is $s(t) = t^4 - 2t^3 + 5$, what is the acceleration $a(t)$ at $t = 2$?

a. 24 ✓

b. 36

c. 12

d. 48

Velocity $v(t) = 4t^3 - 6t^2$. Acceleration $a(t) = 12t^2 - 12t$. At $t = 2$: $a(2) = 12(4) - 12(2) = 48 - 24 = 24$.

32 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

What is the 50th derivative of $f(x) = \cos x$?

a. $-\cos x$ ✓

b. $\cos x$

c. $\sin x$

d. $-\sin x$

Derivatives of cosine repeat in cycles of 4: $f^{(0)} = \cos x$, $f^{(1)} = -\sin x$, $f^{(2)} = -\cos x$, $f^{(3)} = \sin x$. Since $50 = 4(12) + 2$, $f^{(50)}(x) = f^{(2)}(x) = -\cos x$.

33 [Intermediate] • Target: The Product Rule

Use the product rule to find $f'(x)$ if $f(x) = x^2(2x - 1)$.

a. $2x(3x - 1)$ ✓

b. $4x$

c. $6x^2 - 2x - 2$

d. $x(6x - 1)$

Product rule: $f'(x) = 2x^1(2x - 1) + x^2(2) = 2x(3x - 1)$.

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34 [Intermediate] • Target: The Quotient Rule

Use the product rule to find $f'(x)$ if $f(x) = x^2(1x + 3)$.

a. $2x$

b. $x(3x + 7)$

c. $3x(x + 2)$ ✓

d. $3x^2 + 6x - 1$

Product rule: $f'(x) = 2x^1(1x + 3) + x^2(1) = 3x(x + 2)$.

35 [Intermediate] • Target: Derivatives of Basic Trigonometric Functions

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{2x + 0}$.

a. 2

b. $\frac{1}{2x}$

c. 0 ✓

d. 1

Quotient rule: $f'(x) = \frac{1 \cdot (2x + 0) - x \cdot 2}{(2x + 0)^2} = 0$.

36 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Use the chain rule to find $f'(x)$ if $f(x) = (-3x - 3)^4$.

a. $4(-3x - 3)^3$

b. $-12(-3x - 3)^4$

c. $-12(-3x - 3)^3$ ✓

d. $-3(-3x - 3)^3$

Chain rule: $f'(x) = 4(-3x - 3)^3 \cdot -3 = -12(-3x - 3)^3$.

37 [Intermediate] • Target: The Product Rule

Use the chain rule to find $f'(x)$ if $f(x) = (-2x - 5)^2$.

a. $2(-2x - 5)^1$

b. $-4(-2x - 5)^2$

c. $-2(-2x - 5)^1$

d. $-4(-2x - 5)^1$ ✓

Chain rule: $f'(x) = 2(-2x - 5)^1 \cdot -2 = -4(-2x - 5)^1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Quotient Rule • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Use the chain rule to find $f'(x)$ if $f(x) = (-2x - 5)^3$.

a. $3(-2x - 5)^2$

b. $-2(-2x - 5)^2$

c. $-6(-2x - 5)^3$

d. $-6(-2x - 5)^2$ ✓

Chain rule: $f'(x) = 3(-2x - 5)^2 \cdot -2 = -6(-2x - 5)^2$.

39 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use the chain rule to find $f'(x)$ if $f(x) = (-2x - 6)^4$.

a. $-8(-2x - 6)^4$

b. $4(-2x - 6)^3$

c. $-2(-2x - 6)^3$

d. $-8(-2x - 6)^3$ ✓

Chain rule: $f'(x) = 4(-2x - 6)^3 \cdot -2 = -8(-2x - 6)^3$.

40 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use the chain rule to find $f'(x)$ if $f(x) = (-3x + 2)^3$.

a. $-9(-3x + 2)^3$

b. $-9(-3x + 2)^2$ ✓

c. $3(-3x + 2)^2$

d. $-3(-3x + 2)^2$

Chain rule: $f'(x) = 3(-3x + 2)^2 \cdot -3 = -9(-3x + 2)^2$.

41 [Challenging] • Target: The Product Rule • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Use the chain rule to find $f'(x)$ if $f(x) = (-3x + 3)^3$.

a. $3(-3x + 3)^2$

b. $-9(-3x + 3)^2$ ✓

c. $-9(-3x + 3)^3$

d. $-3(-3x + 3)^2$

Chain rule: $f'(x) = 3(-3x + 3)^2 \cdot -3 = -9(-3x + 3)^2$.

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42 [Challenging] • Target: The Quotient Rule • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Use the chain rule to find $f'(x)$ if $f(x) = (-2x - 1)^3$.

a. $3(-2x - 1)^2$

b. $-2(-2x - 1)^2$

c. $-6(-2x - 1)^2$ ✓

d. $-6(-2x - 1)^3$

Chain rule: $f'(x) = 3(-2x - 1)^2 \cdot -2 = -6(-2x - 1)^2$.

43 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Use the chain rule to find $f'(x)$ if $f(x) = (3x - 1)^4$.

a. $4(3x - 1)^3$

b. $12(3x - 1)^3$ ✓

c. $12(3x - 1)^4$

d. $3(3x - 1)^3$

Chain rule: $f'(x) = 4(3x - 1)^3 \cdot 3 = 12(3x - 1)^3$.

44 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 8.01 Area Between Curves

Use the chain rule to find $f'(x)$ if $f(x) = (3x - 2)^4$.

a. $12(3x - 2)^4$

b. $4(3x - 2)^3$

c. $3(3x - 2)^3$

d. $12(3x - 2)^3$ ✓

Chain rule: $f'(x) = 4(3x - 2)^3 \cdot 3 = 12(3x - 2)^3$.

45 [Challenging] • Target: The Product Rule • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Use the chain rule to find $f'(x)$ if $f(x) = (3x + 4)^3$.

a. $9(3x + 4)^3$

b. $9(3x + 4)^4$

c. $9(3x + 4)^2$ ✓

d. $3(3x + 4)^2$

Chain rule: $f'(x) = 3(3x + 4)^2 \cdot 3 = 9(3x + 4)^2$.

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